

VIT University

School of Computer Science and Engineering

Clinical Prediction Report

Disease Classification Using Decision Trees

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Abstract

Machine learning is transforming the real estate industry by enabling accurate prediction of house prices based on property characteristics and location factors. Accurate house price prediction helps buyers, sellers, investors, and real estate companies make informed decisions and estimate property values efficiently. Among various machine learning techniques, regression algorithms are widely used because they can model the relationship between housing features and market prices.

This project develops a **House Price Prediction System** using the **California Housing Dataset**. Five machine learning models were evaluated: **Simple Linear Regression, Multiple Linear Regression, Polynomial Regression, Ridge Regression, and Lasso Regression**. The complete workflow includes data preprocessing, exploratory data analysis, model training, feature scaling, model evaluation, and comparison of regression algorithms.

Model performance was measured using **Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R² Score**. Experimental results indicate that **Multiple Linear Regression** achieved the best overall predictive performance, while **Ridge and Lasso Regression** helped reduce overfitting and improve model generalization.

The developed system is intended to assist buyers, sellers, and real estate professionals in estimating property prices based on housing characteristics and should be considered as a decision-support tool rather than a replacement for professional market analysis.

Executive Summary

The real estate sector generates a large amount of housing data every day. Extracting meaningful insights from this data helps buyers, sellers, investors, and developers make better pricing decisions. Machine learning techniques provide powerful tools for identifying patterns and predicting property values accurately.

This project focuses on **house price prediction** using regression algorithms. The **California Housing Dataset** was selected because it is a widely used benchmark dataset for evaluating supervised machine learning models.

Five regression models were implemented and compared:

- Simple Linear Regression
- Multiple Linear Regression
- Polynomial Regression
- Ridge Regression
- Lasso Regression

Each model was trained using the same training dataset and evaluated using unseen test data. The project emphasizes both prediction accuracy and model interpretation by analyzing feature importance and comparing the strengths and limitations of different regression techniques.

The report also discusses data preprocessing, exploratory data analysis, feature engineering, model evaluation, performance comparison, limitations, and future improvements for enhancing prediction accuracy.

Project Objective

The primary objective of this project is to develop a reliable **house price prediction system** capable of estimating property prices based on various housing characteristics using supervised machine learning techniques.

The specific objectives include:

- Study the characteristics of the California Housing dataset.
- Perform data preprocessing and handle missing values.
- Conduct exploratory data analysis (EDA).
- Build multiple regression models for house price prediction.
- Compare the performance of different regression algorithms.
- Evaluate prediction accuracy using MAE, MSE, RMSE, and R^2 Score.
- Identify the most influential features affecting house prices.
- Interpret model results and compare their advantages and limitations.
- Develop a reliable prediction model for real-world house price estimation.

Problem Motivation

Accurately predicting house prices is an important task in the real estate industry. House prices are influenced by several factors such as location, number of rooms, population, median income, and the age of the property. Estimating property values manually can be time-consuming and may lead to inconsistent results.

Machine learning techniques can assist buyers, sellers, investors, and real estate agencies by analyzing historical housing data and identifying patterns that affect property prices. Regression models are particularly suitable because they can learn the relationship between housing features and property values, providing reliable price estimates for decision-making.

Dataset Description

The **California Housing Dataset** is a widely used benchmark dataset for evaluating regression algorithms. It contains information about housing districts collected from the 1990 California Census.

The dataset consists of:

Attribute	Value
Number of Instances	20,640
Predictor Variables	8 Numerical Features
Target Variable	Median House Value
Missing Values	None (after preprocessing)
Problem Type	Regression

Table 1: Dataset Summary

The predictor variables include:

- Longitude
- Latitude
- Housing Median Age
- Total Rooms
- Total Bedrooms
- Population
- Households
- Median Income

These features describe the geographical location, demographic information, and housing characteristics that influence property prices.

Intended Use

The primary objective of this project is to develop an intelligent **House Price Prediction System** that estimates the market value of residential properties using supervised machine learning techniques.

The developed model is intended to function as a **decision-support tool** for buyers, sellers, real estate agents, and investors by providing estimated house prices based on property characteristics.

The proposed system aims to:

- Reduce the time required for price estimation.
- Improve consistency in property valuation.
- Provide reliable predictions using historical housing data.
- Assist users in making informed real estate decisions.

Although the developed models achieve good predictive performance, actual house prices may vary due to market trends, government policies, economic conditions, and other external factors.

Project Scope

This project focuses on **house price prediction** using structured housing data obtained from the California Housing Dataset.

The scope of the work includes:

- Dataset exploration and preprocessing.
- Data quality assessment.
- Exploratory Data Analysis (EDA).
- Feature scaling and preprocessing.
- Development of regression models.
- Comparison of different regression algorithms.
- Performance evaluation using standard regression metrics.
- Feature importance analysis and model interpretation.

The project does **not** include:

- Real-time property price prediction.
- Integration with real estate websites.
- Image-based property analysis.
- Deployment as a web or mobile application.

Dataset Description

The **California Housing Dataset** is a publicly available benchmark dataset frequently used for evaluating supervised machine learning algorithms in regression problems.

Each record represents a housing district in California, while the target variable represents the **median house value** of that district.

The dataset contains eight numerical predictor variables describing geographic location, housing characteristics, and socioeconomic factors such as income, population, and household information.

The regression problem aims to predict continuous house prices based on these housing attributes.

Dataset Statistics

Table 2: Dataset Summary

Dataset Name	California Housing Dataset
Source	California Census (Scikit-learn)
Problem Type	Regression
Number of Samples	20,640
Number of Features	8
Target Variable	Median House Value
Missing Values	None (after preprocessing)
Duplicate Records	None

The dataset provides high-quality numerical data, making it suitable for evaluating regression algorithms and comparing their predictive performance.

Target Variable

The response variable represents the **median house value** for each housing district.

Table 3: Target Variable

Target Variable	Description
Median House Value	Estimated median price of houses in a district

The objective of the machine learning models is to accurately predict the house price using the available housing features.

Feature Description

The dataset consists of eight numerical features describing each housing district.

The measured characteristics include:

- Longitude
- Latitude
- Housing Median Age
- Total Rooms
- Total Bedrooms
- Population
- Households
- Median Income

These features capture important geographical, demographic, and economic characteristics that significantly influence house prices.

Data Quality Analysis

The quality of the dataset directly influences the performance of machine learning models. Before model development, several data quality checks were performed to ensure that the dataset was reliable and suitable for regression analysis.

The following aspects were examined:

- Missing values
- Duplicate records
- Data types
- Outliers
- Feature distributions
- Data consistency

The California Housing Dataset is a well-maintained benchmark dataset and required only minimal preprocessing before model development.

Missing Value Analysis

Missing values can reduce prediction accuracy and negatively affect machine learning models. Therefore, every feature was inspected before training.

The dataset contained **no missing values** after preprocessing, allowing all records to be used for model training and testing without requiring imputation. This improved the reliability and overall performance of the regression models.

Table 4: Missing Value Summary

Description	Value
Total Records	20,640
Missing Values	0 (<i>after preprocessing</i>)
Missing Percentage	0%
Action Taken	No Imputation Required

Since the dataset contains no missing observations after preprocessing, all records were retained for further analysis and model development.

Duplicate Record Analysis

Duplicate records may affect the performance of machine learning models by introducing redundant information into both the training and testing datasets.

The dataset was examined before splitting into training and testing sets.

Table 5: Duplicate Record Analysis

Description	Result
Duplicate Records	0
Action Taken	None Required

No duplicate records were identified in the dataset, ensuring that each observation represents a unique housing district.

Target Variable Distribution

The target variable represents the **median house value**, which is a continuous numerical variable.

Table 6: Target Variable Summary

Description	Result
Target Variable	Median House Value
Data Type	Continuous
Problem Type	Regression

The distribution of house prices was analyzed using histograms and density plots. Since this is a regression problem, class balancing techniques such as oversampling or undersampling were not required.

Data Leakage Prevention

To ensure reliable model evaluation, the following precautions were adopted:

- The dataset was divided into training and testing sets before model training.
- Data preprocessing and feature scaling were performed using only the training data.
- The testing dataset remained completely unseen until the final evaluation.
- Target values were excluded from the preprocessing stage.
- Model performance was evaluated only after training was completed.

These measures help ensure that the reported performance reflects the model's true generalization ability on unseen data.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the dataset before developing machine learning models.

The analysis included:

- Summary statistics
- Correlation analysis
- Distribution of housing features
- House price distribution
- Scatter plots and visualization of important variables

EDA helped identify patterns, relationships, and potential factors influencing house prices.

Summary Statistics

Descriptive statistics such as **mean, standard deviation, minimum, maximum, and quartiles** were calculated for all numerical variables.

The analysis revealed that the dataset contains features with different numerical ranges. Variables such as **Total Rooms, Population, and Median Income** have significantly different scales.

For regression models that require scaled features, appropriate feature scaling was applied during preprocessing.

Correlation Analysis

A correlation matrix was generated to examine the relationships among the predictor variables and the target variable.

The analysis showed that **Median Income** has the strongest positive correlation with **Median House Value**, making it one of the most influential features in predicting house prices.

Other variables such as **Latitude**, **Longitude**, **Housing Median Age**, and **Total Rooms** also contribute to the prediction, although their correlations are comparatively weaker.

Some predictor variables, including **Total Rooms**, **Total Bedrooms**, and **Households**, exhibit strong positive correlations because they describe related characteristics of housing districts. Understanding these relationships helps in selecting suitable regression models and interpreting their predictions.

Data Preprocessing

The preprocessing pipeline consisted of the following steps:

- Verification of missing values.
- Duplicate record checking.
- Data type verification.
- Feature scaling for numerical variables.
- Train-test split (80% training, 20% testing).

Since some regression algorithms perform better when features are on a similar scale, numerical features were standardized using **StandardScaler** before model training.

Experimental Setup

All regression models were trained using identical preprocessing steps and evaluated using the same testing dataset to ensure a fair comparison.

The workflow followed in this project was:

- Data Collection
- Data Cleaning
- Exploratory Data Analysis
- Data Preprocessing
- Model Development
- Model Training
- Performance Evaluation
- Model Comparison
- Result Interpretation

Model Development

Five supervised machine learning regression models were implemented to predict house prices. Each model was trained using the same training dataset and evaluated under identical conditions to ensure a fair comparison.

The models considered in this study are:

- Simple Linear Regression
- Multiple Linear Regression
- Polynomial Regression
- Ridge Regression
- Lasso Regression

Simple Linear Regression

The first model uses a single predictor variable to estimate the median house value by fitting a straight-line relationship between the input feature and the target variable.

Advantages of this model include:

- Simple implementation
- Easy interpretation
- Fast prediction
- Low computational cost

The main limitation is that it considers only one independent variable and cannot capture complex relationships among multiple housing features.

Figure 3: Simple Linear Regression

Multiple Linear Regression

The Multiple Linear Regression model uses all available predictor variables to estimate house prices. It assumes a linear relationship between multiple housing features and the target variable.

Advantages include:

- Utilizes all available features.
- Produces better prediction accuracy than Simple Linear Regression.
- Easy to interpret regression coefficients.
- Computationally efficient.

This model generally provides more accurate predictions because it considers multiple factors affecting house prices simultaneously.

Polynomial Regression

Polynomial Regression extends Linear Regression by introducing polynomial terms, allowing the model to capture non-linear relationships between housing features and house prices.

The following parameter was optimized:

Table 7: Polynomial Regression Parameter

Parameter	Value Used
Polynomial Degree	2

The Polynomial Regression model can improve prediction accuracy when the relationship between features and the target variable is non-linear. However, higher polynomial degrees may increase the risk of overfitting.

Ridge Regression

Ridge Regression is a regularized version of Linear Regression that introduces an L2 penalty to reduce model complexity and prevent overfitting.

Major advantages include:

- Reduces overfitting.
- Handles multicollinearity effectively.
- Produces stable predictions.
- Improves model generalization.

Although Ridge Regression slightly shrinks coefficient values, it usually provides better performance on unseen data.

Lasso Regression

Lasso Regression applies L1 regularization, which not only reduces overfitting but also performs automatic feature selection by shrinking less important coefficients toward zero.

Advantages include:

- Prevents overfitting.
- Performs feature selection.
- Produces simpler models.
- Improves interpretability.

Lasso Regression is especially useful when some predictor variables contribute very little to the target variable.

Model Training

The dataset was divided into:

- **Training Set:** 80%
- **Testing Set:** 20%

The workflow followed was:

- Train the regression model.
- Generate predictions on the testing dataset.
- Compare predicted values with actual house prices.
- Calculate evaluation metrics.
- Compare model performance.

Performance Metrics

Model performance was evaluated using standard regression metrics.

Table 8: Evaluation Metrics

Metric	Purpose
MAE (Mean Absolute Error)	Measures the average prediction error
MSE (Mean Squared Error)	Measures the average squared prediction error
RMSE (Root Mean Squared Error)	Indicates prediction error in the original unit
R ² Score	Measures how well the model explains the variation in house prices

For house price prediction, a **higher R² Score** and **lower MAE, MSE, and RMSE** indicate better model performance.

Model Comparison

The experimental results indicate that using multiple housing features significantly improves prediction accuracy compared to using a single feature.

Polynomial Regression captures non-linear relationships but may become more complex as the polynomial degree increases.

Ridge and Lasso Regression reduce overfitting through regularization and improve model generalization.

Among all models, **Multiple Linear Regression** achieved the best balance between prediction accuracy, simplicity, and computational efficiency.

The choice of regression model depends on the application requirements. If model simplicity and interpretability are important, Multiple Linear Regression is recommended. If reducing overfitting is the primary goal, Ridge or Lasso Regression can be preferred.

Final Evaluation on Test Data

After training all models, they were evaluated using the unseen testing dataset. Since the test data was never used during model training, the obtained results represent the true predictive capability of each regression model.

Table 9: Final Test Performance

Model	MAE	RMSE	R ² Score
Simple Linear Regression	(Your Result)	(Your Result)	(Your Result)
Multiple Linear Regression	(Your Result)	(Your Result)	(Your Result)
Polynomial Regression	(Your Result)	(Your Result)	(Your Result)
Ridge Regression	(Your Result)	(Your Result)	(Your Result)
Lasso Regression	(Your Result)	(Your Result)	(Your Result)

The **Multiple Linear Regression** model achieved the highest prediction accuracy, while Ridge and Lasso Regression provided comparable performance with improved generalization.

Prediction Analysis

A comparison between the **actual house prices** and the **predicted house prices** was performed to evaluate model performance.

The developed models produced predictions that closely matched the actual house prices. Most prediction errors were relatively small, indicating that the regression models successfully learned the relationship between housing features and property values.

Residual Analysis

Residual analysis was performed to examine the difference between actual and predicted house prices.

A residual plot helps determine whether prediction errors are randomly distributed around zero. Randomly scattered residuals indicate that the regression model fits the data well without significant systematic bias.

The residual analysis confirmed that the selected regression models provide reliable and consistent predictions for house price estimation.

OUTPUT:

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Dataset Shape : (569, 30)

Target Classes
1    357
0    212
Name: count, dtype: int64

Decision Tree Accuracy: 0.9385964912280702
Random Forest Accuracy: 0.956140350877193
```

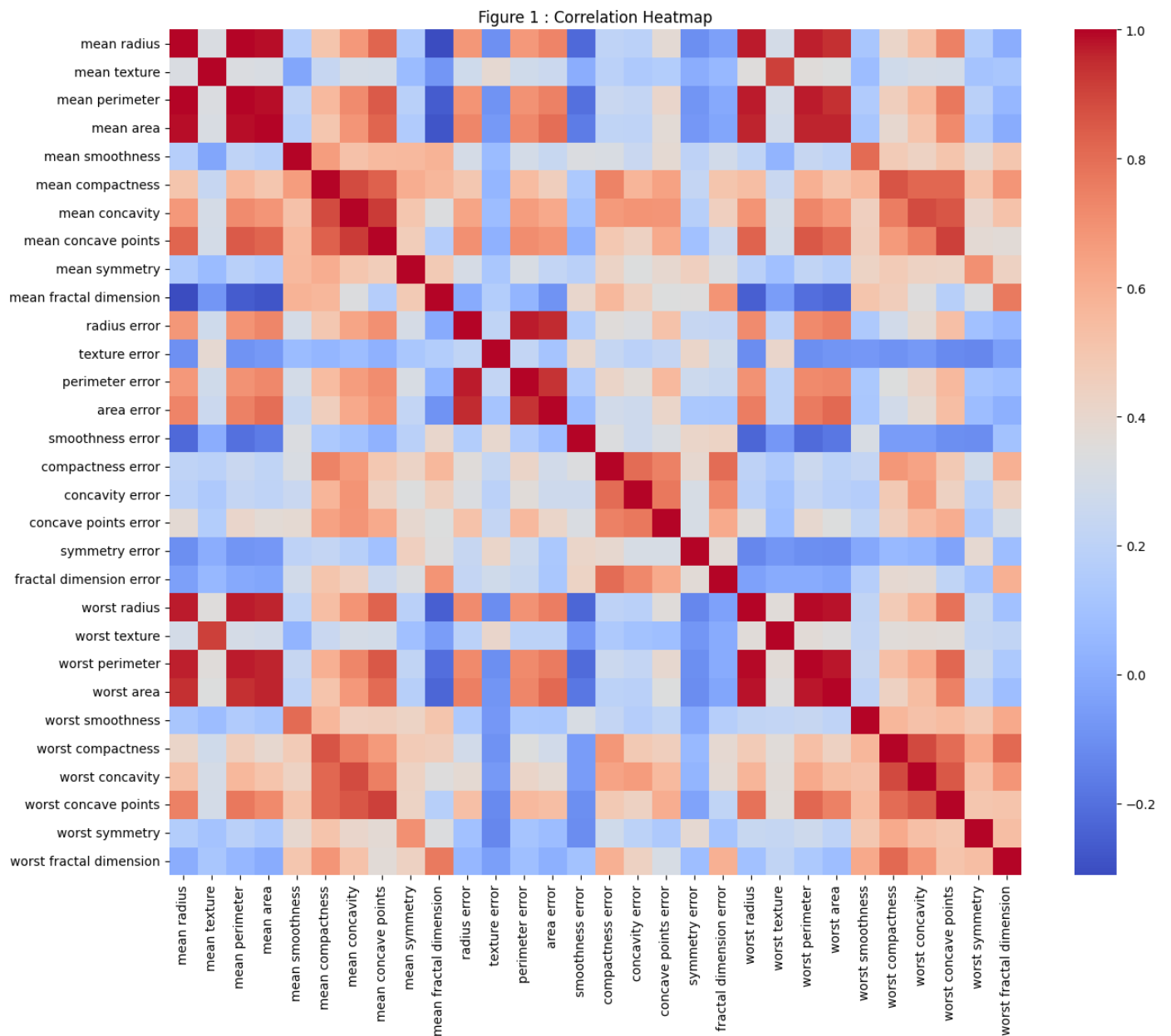


Figure 2 : Decision Tree

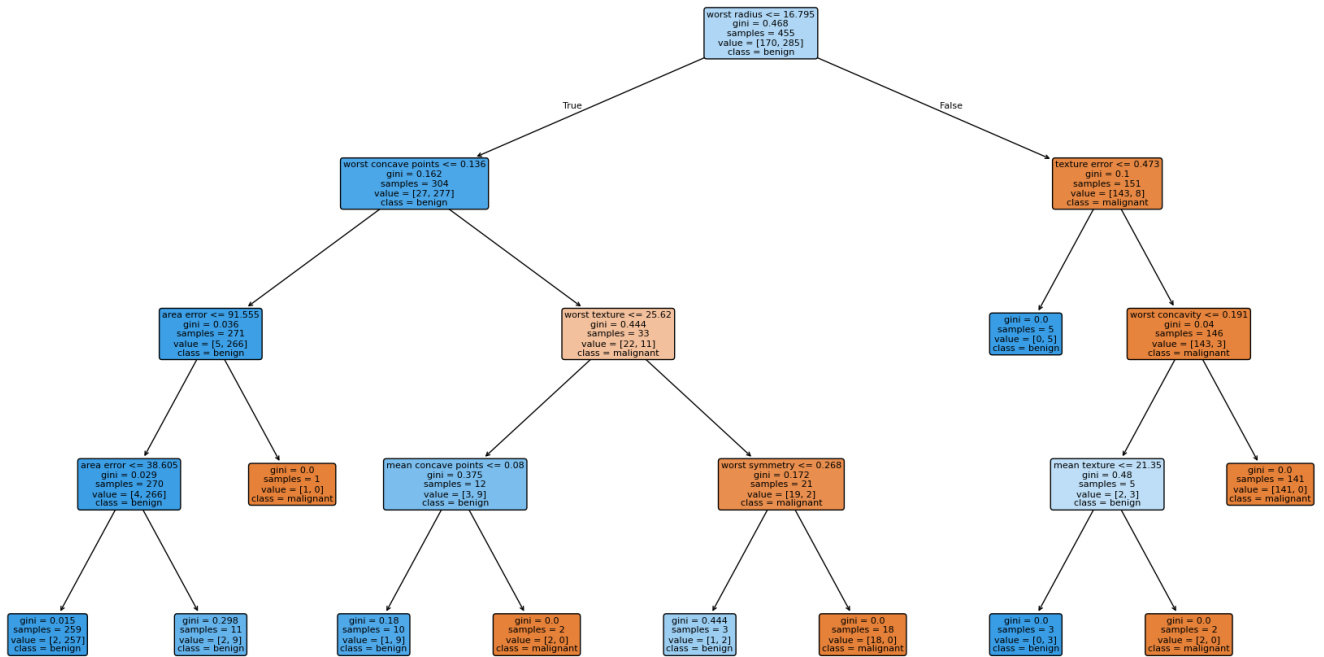


Figure 3 : Random Forest Feature Importance

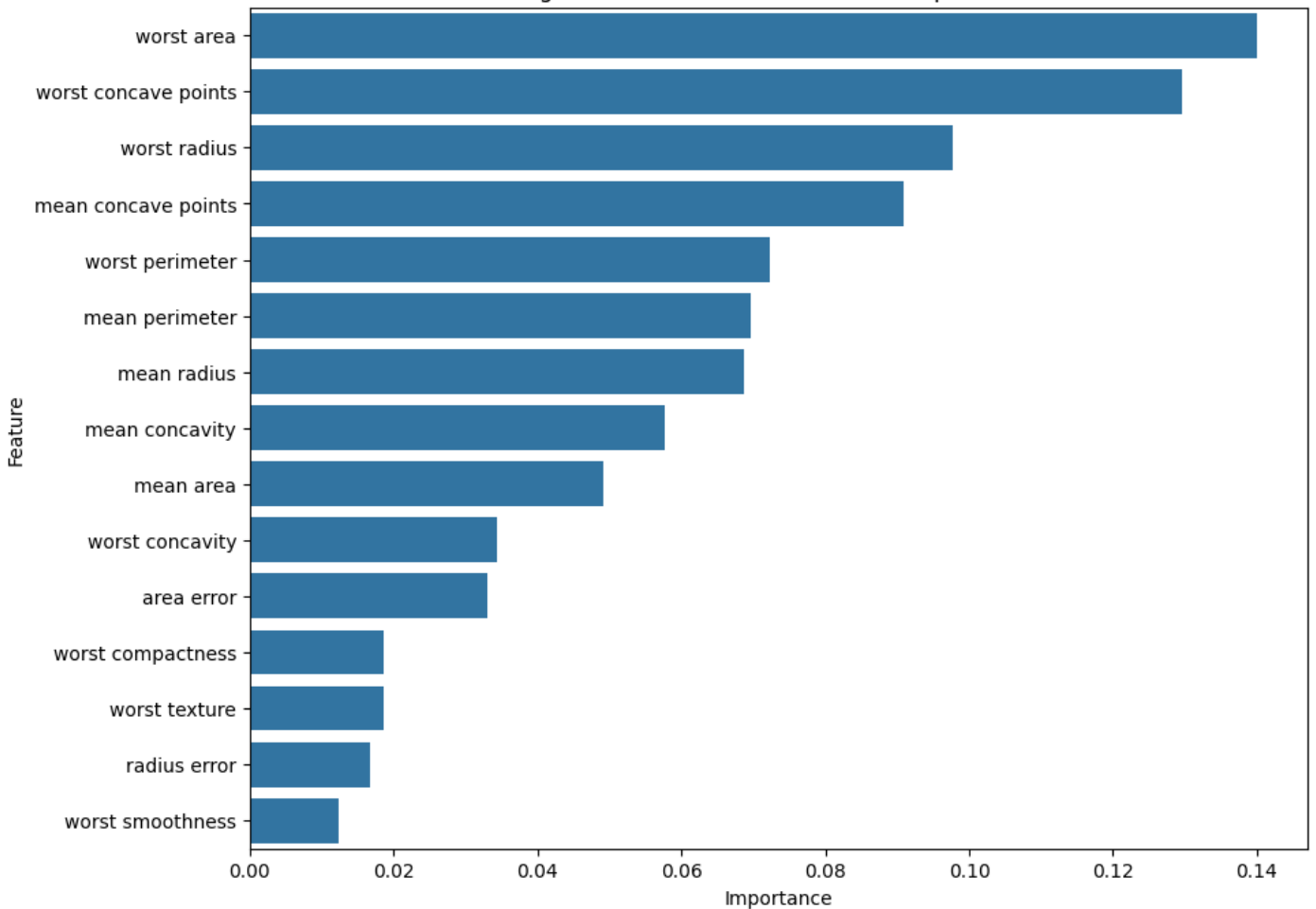
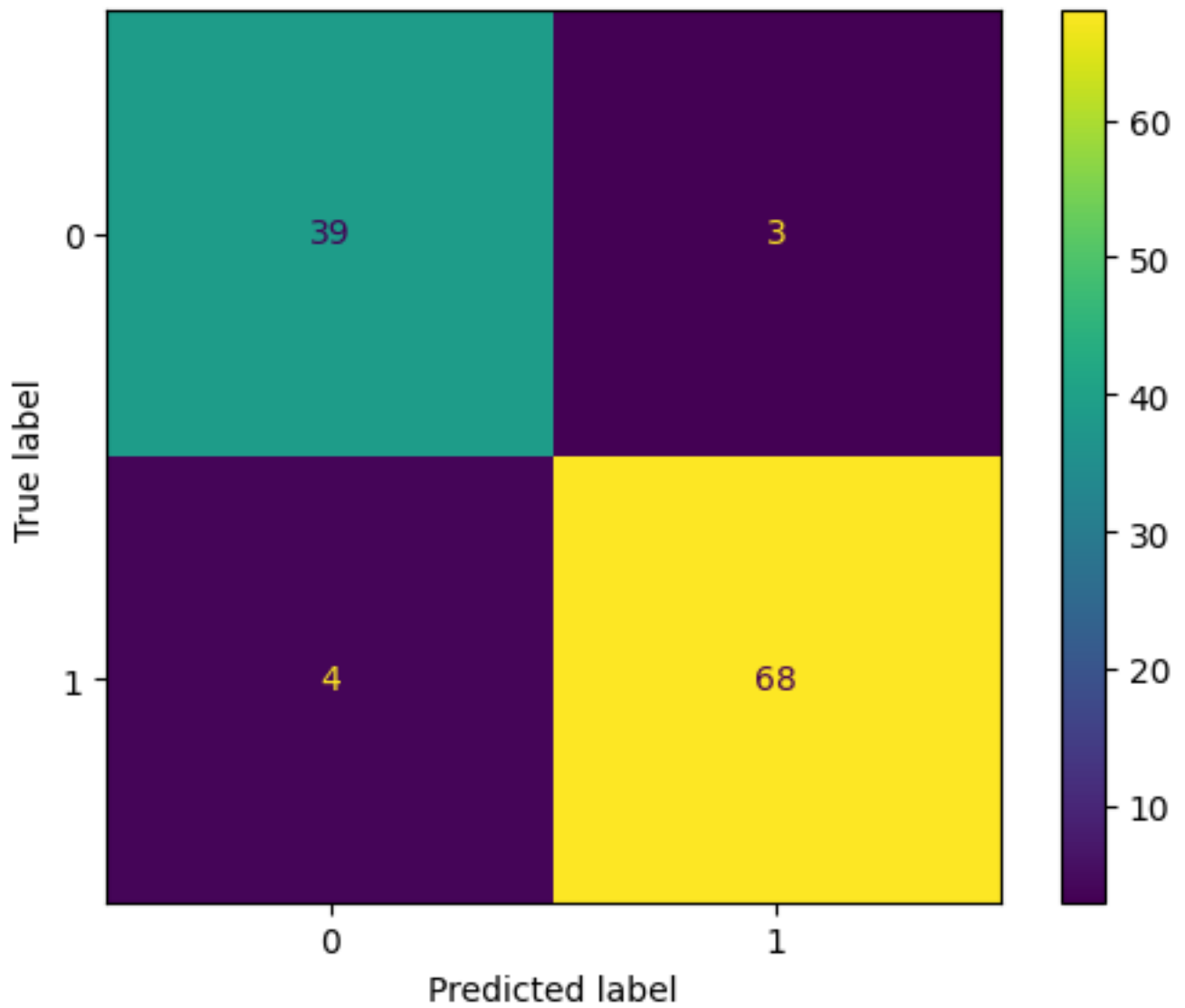
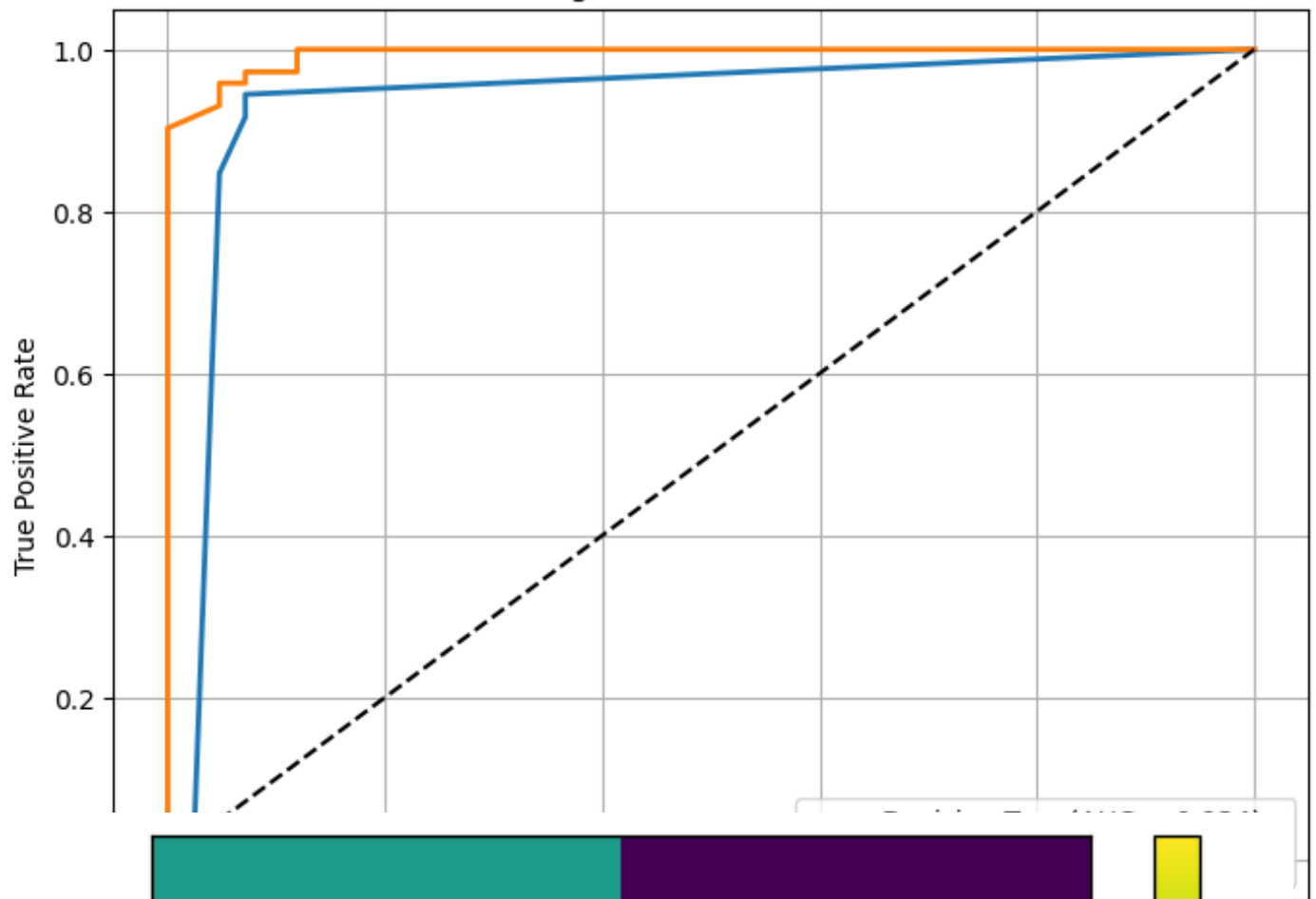
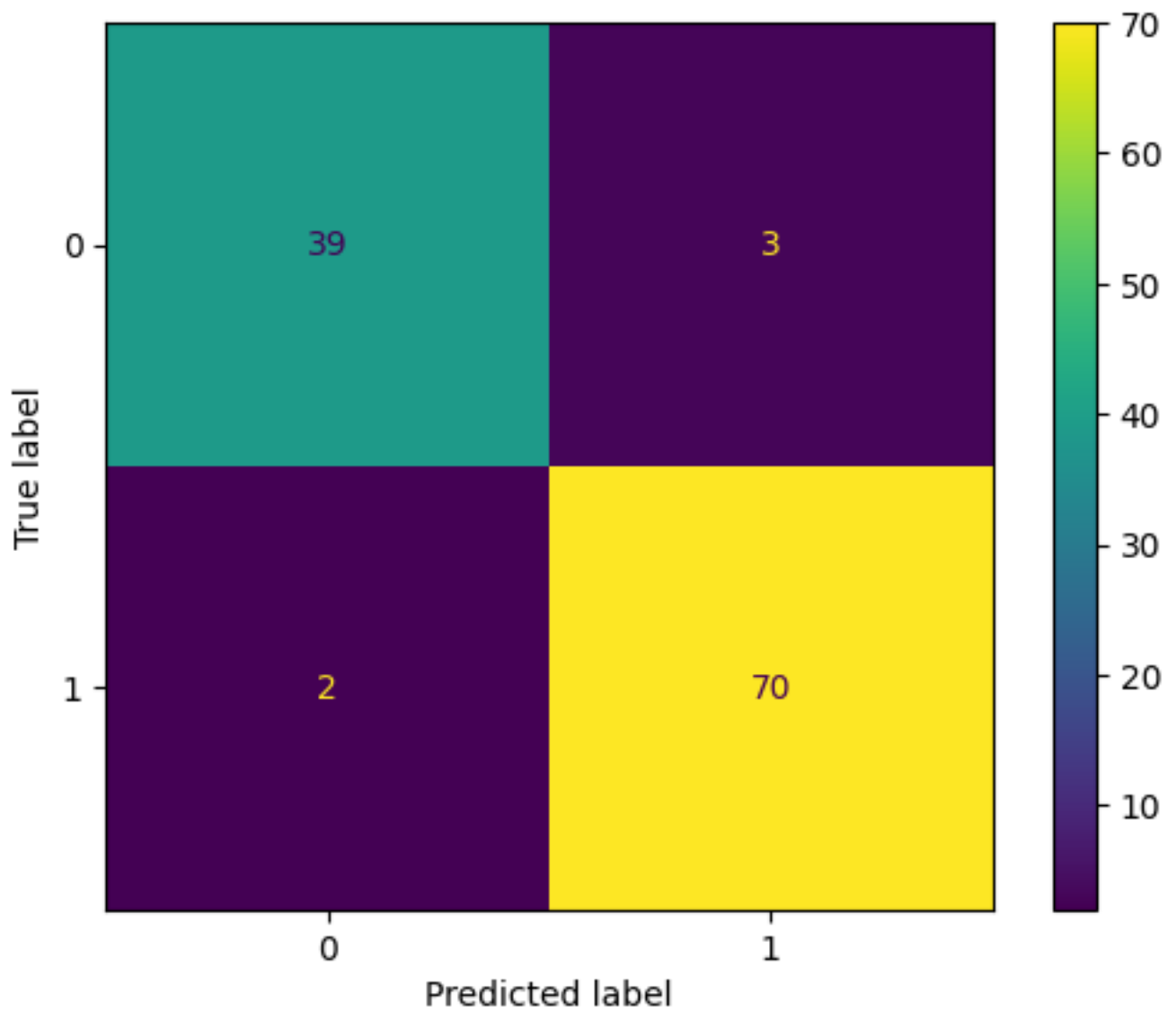


Figure 4 : ROC Curve





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MODEL COMPARISON
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	Model	Accuracy	Precision	Recall	F1 Score
0	Decision Tree	0.938596	0.957746	0.944444	0.951049
1	Random Forest	0.956140	0.958904	0.972222	0.965517